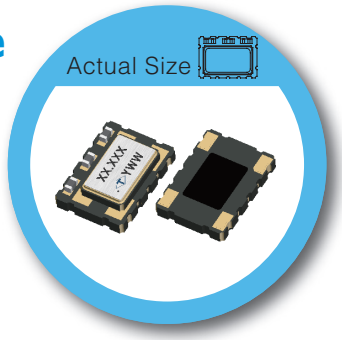
TA Type

7.0 x 5.0 mm SMD Voltage Controlled Temperature Compensated Crystal Oscillator



RoHS Compliant

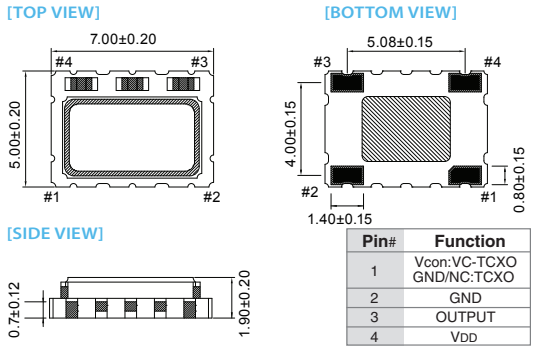
FEATURE

- Typical 7.0 x 5.0 x 1.9 mm ceramic SMD package.
- For automatic assembly.
- Compactness and light weight.
- Low power consumption.
- VCTCXO available.

TYPICAL APPLICATION

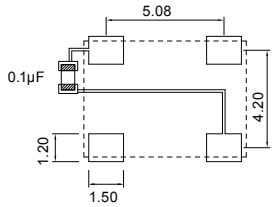
- Femtocell , Base Stations
- WLAN/WiMAX/WIFI, Wireless Communications
- Mobile Phone

DIMENSION (mm)



Pin#	Function
1	Vcon:VC-TCXO GND/NC:TCXO
2	GND
3	OUTPUT
4	VDD

SOLDER PAD LAYOUT (mm)



To ensure optimal oscillator performance, place a by-pass capacitor of 0.1μF as close to the part as possible between Vdd and GND pads.

ELECTRICAL SPECIFICATION

Parameter	3.3 / 3.0 V		2.5 V		Unit
	Min.	Max.	Min.	Max.	
Supply Voltage Variation (VDD)	VDD-5%	VDD+5%	VDD-5%	VDD+5%	V
Frequency Range	5	52	5	52	MHz
Standard Frequency (for Clipped Sine Wave)	8, 8.192, 10, 12.5, 12.8, 16, 16.384, 19.44, 25, 26				
Frequency Tolerance*	–	±2.0	–	±2.0	ppm
Frequency stability					
Vs Supply Voltage (±5%) change	–	±0.1	–	±0.1	ppm
Vs Load (±10%) change	–	±0.2	–	±0.2	
Vs Aging	–	±1.0	–	±1.0	
Supply Current (Clipped Sine Wave)	–	3.5	–	3.5	mA
Output (Clipped Sine Wave)	0.8	–	0.8	–	Vp-p
Load (Clipped Sine Wave)	10 KΩ // 10pF		10 KΩ // 10pF		
Control Voltage Range (VCTCXO)	0.5	2.5	0.4	2.4	V
Pulling Range (VCTCXO)	±5.0	±12.0	±5.0	±12.0	ppm
Vc Input Impedance (VCTCXO)	100	–	100	–	kΩ
Phase Noise @ 19.2 MHz 100 Hz	-115		-115		dBc / Hz
1 kHz	-135		-135		
10 kHz	-148		-148		
Start Time	–	2	–	2	mSec
Storage Temp. Range	-55	125	-55	125	°C

Standard frequencies are frequencies which the crystal has been designed and does not imply a stock position.
*Frequency at 25°C, 1 hour after reflow.

FREQ. STABILITY vs. TEMP. RANGE

Temp. (°C) \ ppm	±0.5	±1.0
-20 ~ +70	○	○
-30 ~ +85	○	○
-40 ~ +85	△"	○

* ○: Available △:Conditional X: Not available
" 10~26MHz and Pulling <8ppm available

Note: not all combination of options are available. Other specifications may be available upon request.